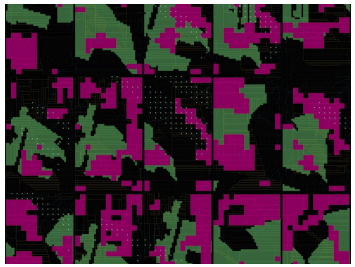
PORTFOLIO · CODE & AI

JOHN NORTHRUP

Architectural designer who draws it, builds it, and automates it.

Permit drawing sets for city and county approval, fabrication from concept to install, and Rhino, Grasshopper, and Python tools that speed up design work.

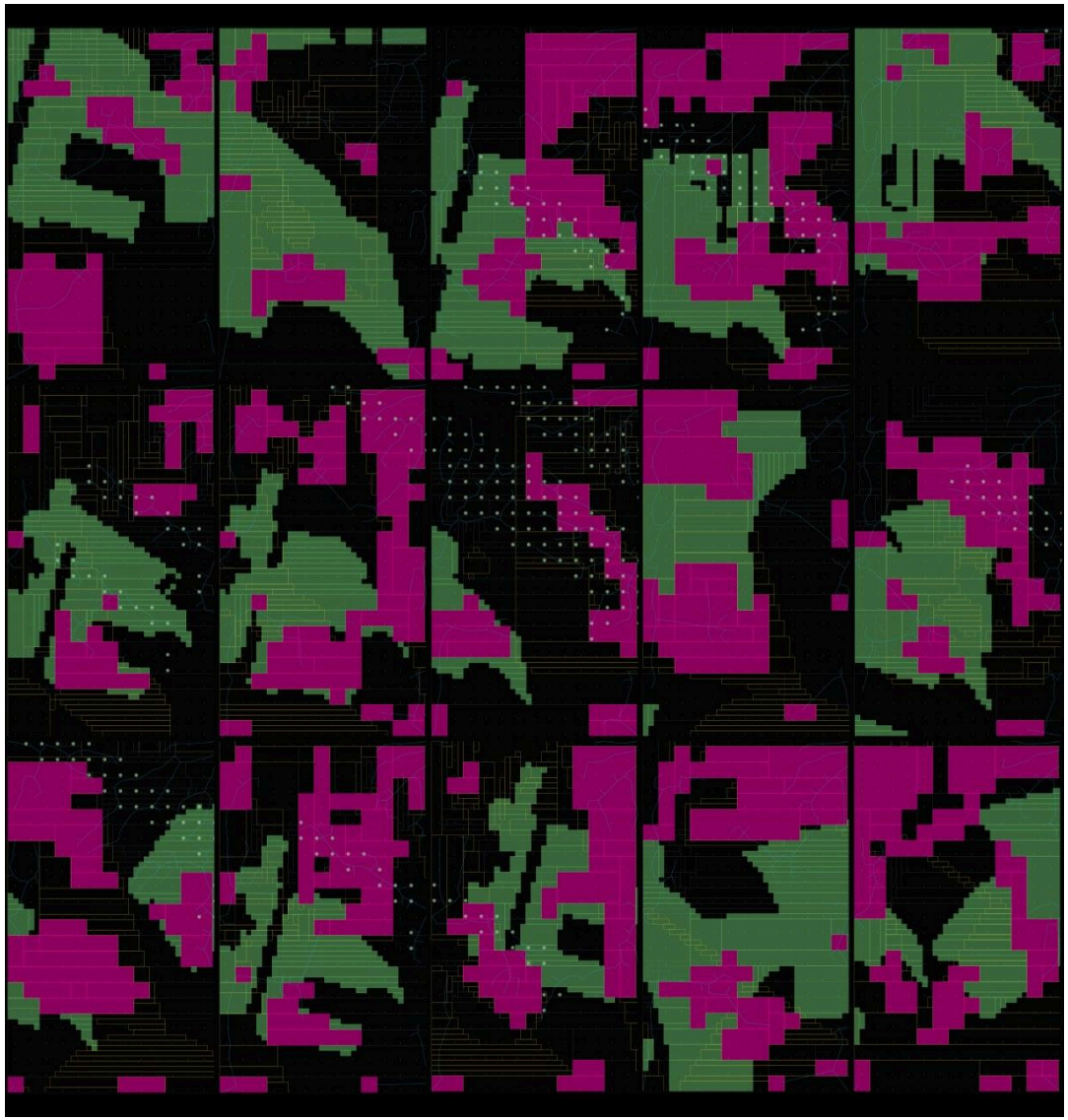




JCNORTHRUP@GMAIL.COM JCNORTHRUP.COM

CODE & AI

01	memoryMachine	SU26, M.ARCH THESIS	03
02	PROTOTYPE	SP26, PROTOTYPE, 3GB STUDIO	06
03	INTERFACE ARCHITECTURE	3GA STUDIO	09



01

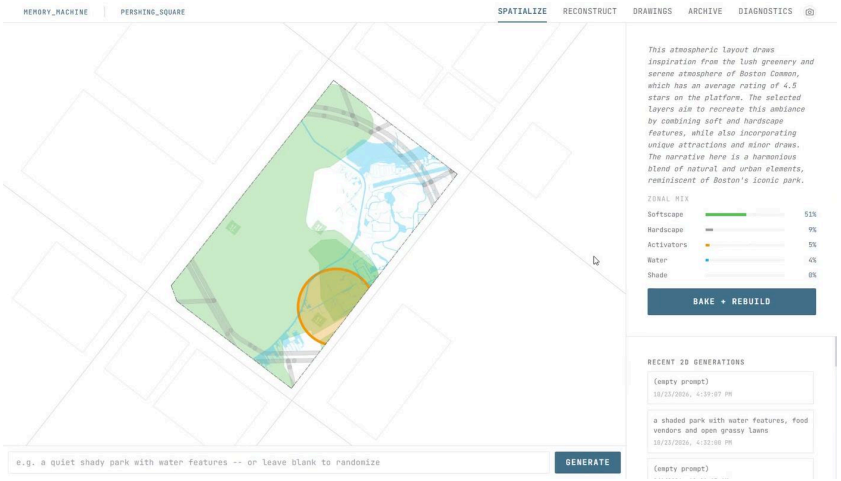
memoryMachine

SU26, M.ARCH THESIS
ADVISOR · CASEY REHM

memoryMachine is a computational design framework that transforms civic sentiment and archival memory into operative spatial data to revitalize under-utilized public space. Using Pershing Square in Downtown Los Angeles as a test bed—a civic site defined by a history of top-down, disconnected redesigns—the project proposes an adaptive spatial infrastructure in place of the fixed, one-time master plan.

At its core, a multimodal data ingestion engine aggregates web reviews, archival records, and spatial metrics, pairing qualitative sentiment with physical variables such as microclimate, canopy coverage, and circulation flows. To ensure equitable representation, the engine applies demographic weighting to mitigate digital access bias. These inputs are translated into spatial density fields and programmatic adjacencies, generating interventions—such as terraced landforms, water systems, and subterranean transit connections—as evidence-based spatial responses rather than predetermined formal gestures.

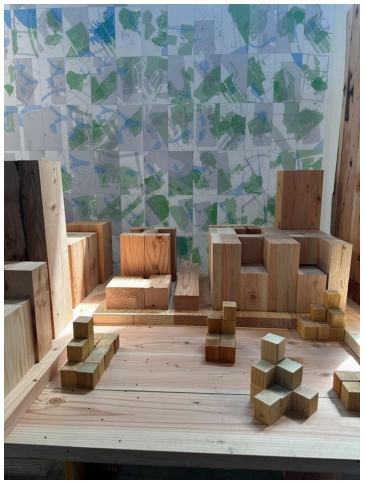
By merging expert analysis with experiential civic data, memoryMachine repositions the architect from sole author to system mediator. Public input shifts from a one-off consultation phase into a permanent, operative design parameter. The result is a self-regulating spatial framework that ingests usage patterns to continuously re-evaluate the site and adapt through phased physical interventions—establishing a democratic model in which public space evolves alongside the community it serves.



SOFTWARE | GENERATE A PLAN, REBUILD IT AS MASSING, DRAW IT, CHECK THE NUMBERS (SP...



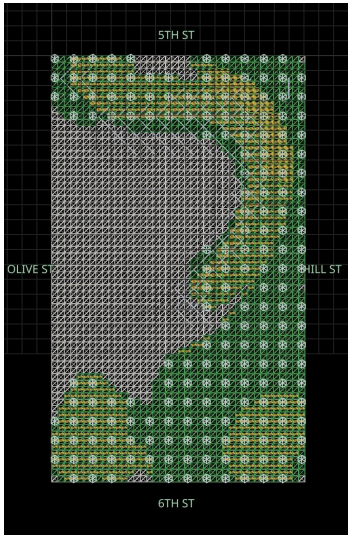
INSTALLATION | TIMBER SITE MODEL IN THE EXHIBITION HALL



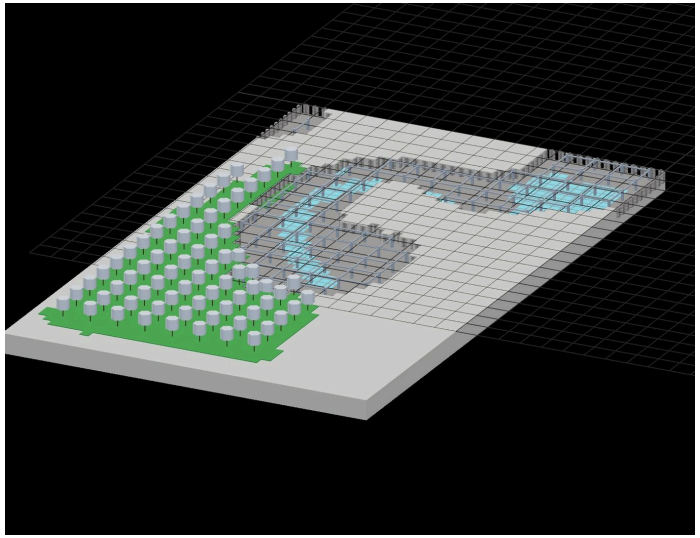
INSTALLATION | TIMBER SITE MODEL A...



RENDER | THE GENERATED PARK IN ITS DOWNTOWN CONTEXT



GENERATED PLAN | PERSHING SQUAR...



GENERATED MODEL | THE SAME SCHEME AS A MASSING



DRAWING | PRECEDENT LIBRARY DRAWN BY THE SYSTEM: PARC DE LA VILLETTE



02

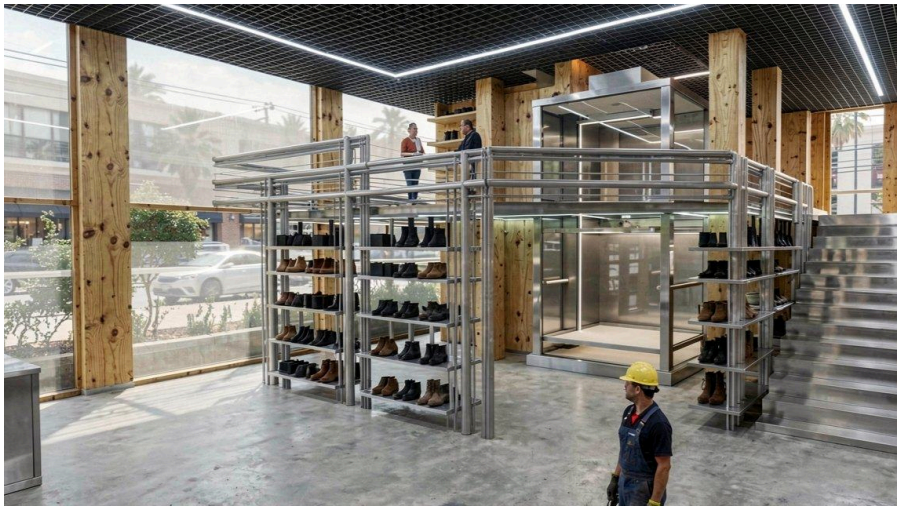
PROTOTYPE

SP26, PROTOTYPE, 3GB STUDIO

A generative pipeline for retail architecture and product design, built around a footwear brand's flagship storefront. Architectural iterations were produced through a custom Flux/ComfyUI pipeline, while the product line — a technical boot shown here in cutaway and colorway studies — was developed in parallel through a Gemini-based image pipeline. The two threads feed each other: the building's industrial material palette of raw wood, steel, and concrete is carried directly into the product's own construction and display.



CUTAWAY SECTION ELEVATION OF THE FULL BUILDING



INTERIOR RETAIL FLOOR WITH STEEL AND TIMBER SHELVING DISPLAY



INTERIOR COUNTER SCENE WITH STAFF AND CUSTOMER



TECHNICAL BOOT PRODUCT RENDER, STUDIO SHOT



BOOT COLORWAY COMPARISON SHEET, FOUR VARIANTS



TECHNICAL CUTAWAY ILLUSTRATION OF THE BOOT'S INTERNAL CO...

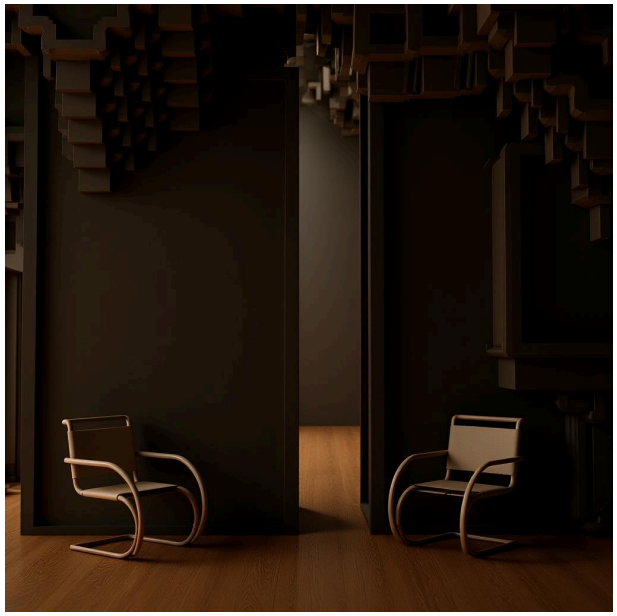


03

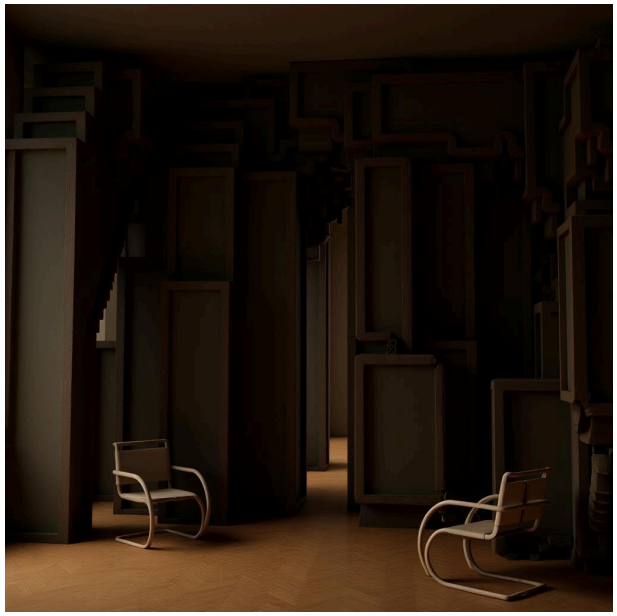
INTERFACE ARCHITECTURE

3GA STUDIO
ADVISOR · CASEY REHM

A Cabinet of Curiosities that produces memory artifacts. The installation receives audio input from the surrounding space; language becomes image through translation of an LLM to 3D object creation via API calls (Meshy). Objects are automatically optimized for 3D printing — oriented, scaled, and combined through a Python pipeline.



TWO SYMMETRIC DARK PORTALS, EACH WITH A SINGLE CHAIR



HALLWAY OF PIXELATED CORNICE PEDESTALS WITH TWO CHAIRS O...



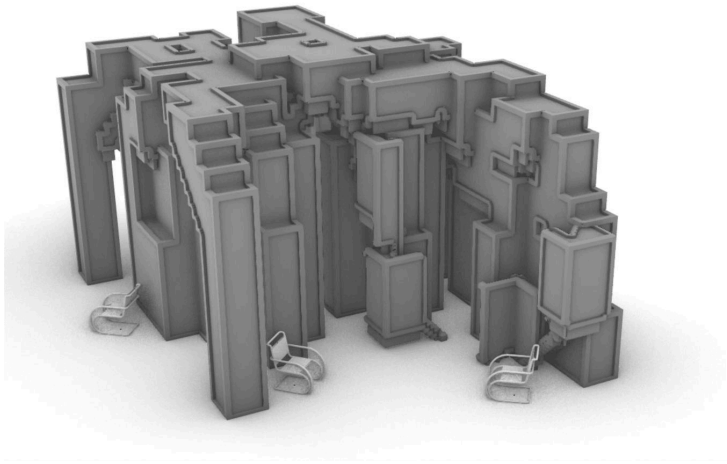
SINGLE CHAIR AGAINST A PIXELATED WOOD CORNICE WALL, DOOR...



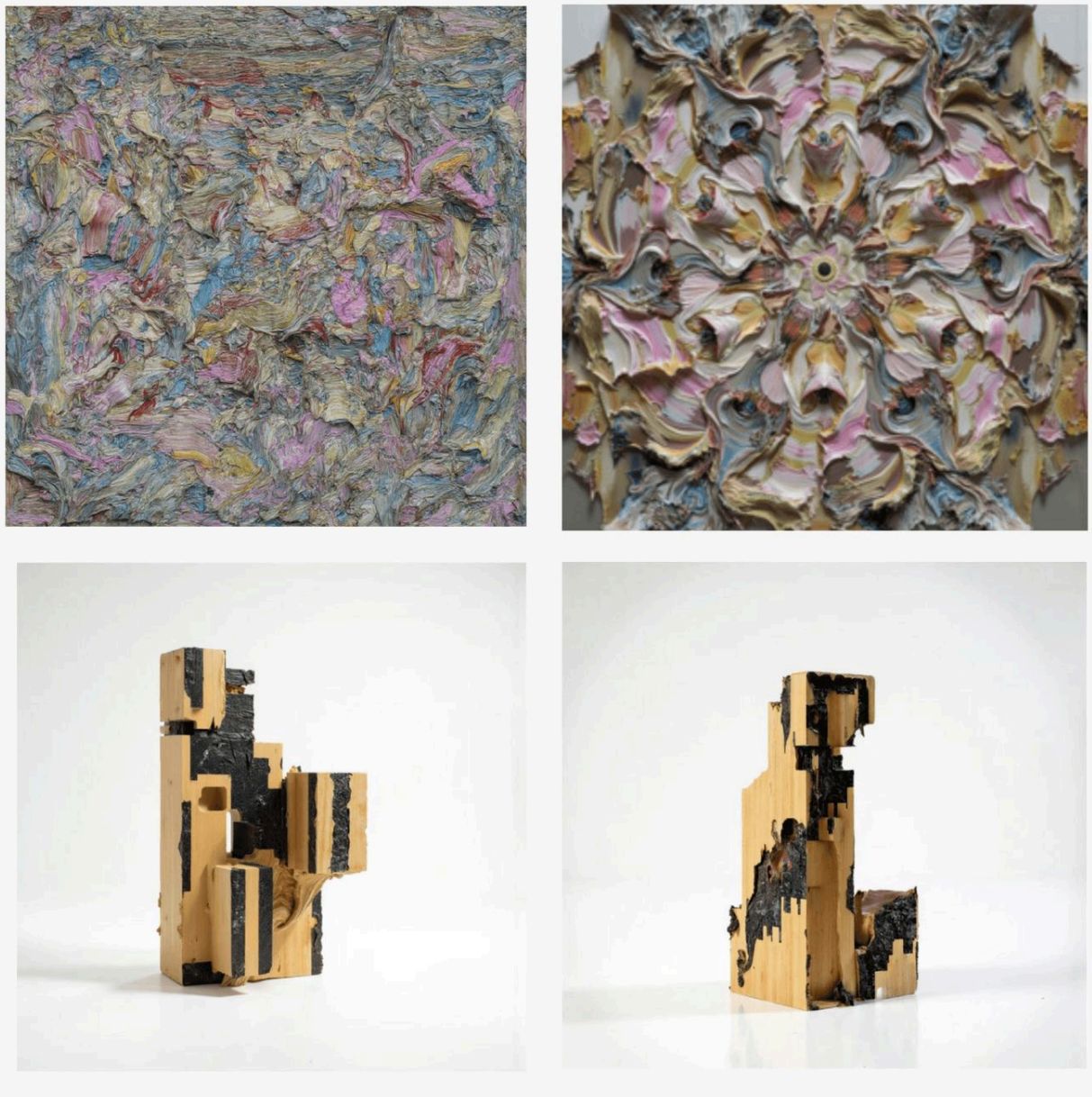
CABINET OF CURIOSITIES INSTALLATION, CROPPED DETAIL VIEW



ELEVATION DRAWING RENDER OF THE CABINET



ASSEMBLED 3D-PRINTED OBJECT RENDER, MASSING VIEW



TEXTURED 3D-PRINTED SCULPTURES, TWO VIEWS

OBJECT STUDIES



BURNT WOOD MEMORY ARTIFACT, STUDY 1



BURNT WOOD MEMORY ARTIFACT, STUDY 2



BURNT WOOD MEMORY ARTIFACT, STUDY 3



BURNT WOOD MEMORY ARTIFACT, STUDY 4

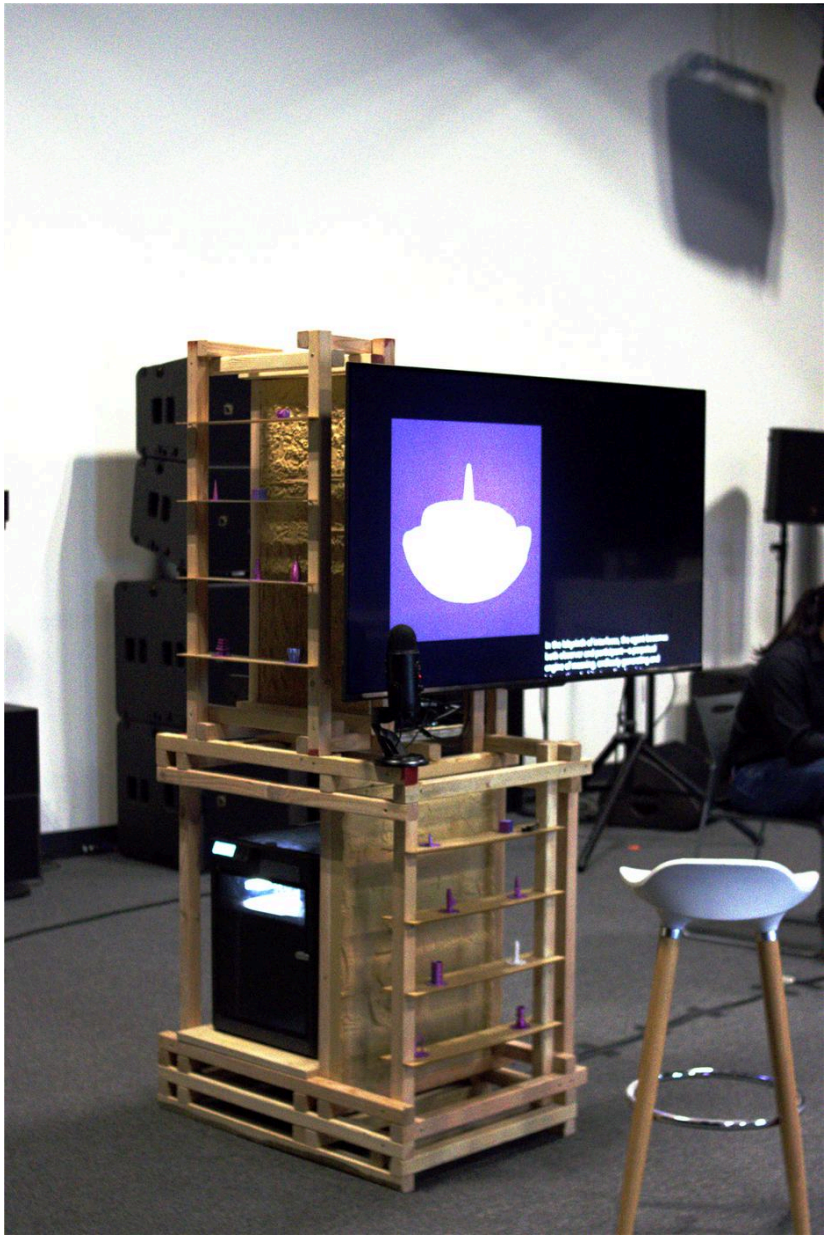


BURNT WOOD MEMORY ARTIFACT, STUDY 5



BURNT WOOD MEMORY ARTIFACT, STUDY 6

OBJECT STUDIES



FINAL INTERACTIVE INSTALLATION WITH SCREEN AND SPEAKER RIG